Alkane

101. The final product of complete oxidation of hydrocarbons is

(a) Acid

(b) Aldehyde

(c) $H_2O + CO_2$

(d) Dihydric alcohol

102. Which of the following will have least hindered rotation about carbon-carbon bond

(a) Ethane

(b) Ethylene

(c) Acetylene

(d) Hexachloroethane

103. Which of the following represents the most oxidized form of hydrocarbon

(a) CO_2 (b) $RCHO$

(c) $RCOOH$ (d) *RCOOOH*

104. Name the reaction

$$C_{10}H_{22} \xrightarrow{900K} C_4H_8 + C_6H_{14}$$

(a) Alkylation

(b) Cracking

(c) Pyrolysis

(d) Fractionation

105. How many types of carbon atoms are present in 2, 2, 3-trimethylpentane

(a) One (b) Two

(c) Three (d) Four

106. Which one gives only one monosubstitution product on chlorination

(a) *n*-pentane

(b) Neopentane

(c) Isopentane

(d) *n*-butane

107. CH_4 is formed when

(a) Sodium acetate is heated with sodalime

(b) Iodomethane is reduced

(c) Aluminium carbide reacts with water

(d) All of these

108. A mixture of methane, ethylene and acetylene gases is passed through a Wolf's bottle containing ammoniacal cuprous chloride. The gas coming out is

(a) Methane

(b) Acetylene

(c) A mixture of methane and ethylene

(d) The original mixture

109. At room temperature solid paraffin is

(a) C_3H_8 (b) C_8H_{18}

(c) C_4H_{10} (d) $C_{20}H_{42}$

110. Which one of the following compounds does not give addition reactions

(a) Aldehydes (b) Alkanes

(c) Alkenes (d) Alkynes

(e) Ketones (f) All of these

111. The most important method of preparation of hydrocarbons of lower carbon number is

(a) Pyrolysis of higher carbon number hydrocarbons

(b) Electrolysis of salts of fatty acids

(c) Sabatier and Senderen's reaction

(d) Direct synthesis

112. The inorganic origin of petroleum is indicated by the fact that

(a) Its constituents can be separated by fractional distillation

(b) Carbon and hydrocarbon can combine by absorption of solar energy to give hydrocarbons

(c) Petroleum contains traces of chlorophyll

(d) Oil fields are located with the help of seismograph

113. Which of the following is a gemdihalide

(a) $CH_3.CHBr.CHBr.CH_3$

(b) $CH_2Br.CH_2Br$

(c) $CHBr = CHBr$

(d) CH_3CHBr_2

114. Which one of the following contain isopropyl group

(a) 2,2,3,3-tetramethylpentane

(b) 2-methylpentane

(c) 2,2,3-trimethylpentane

(d) 3,3-dimethylpentane

115. Natural gas is a mixture of

(a) $CO + CO_2$

(b) $CO + N_2$

(c) $CO + H_2 + CH_4$

(d) $CH_4 + C_2H_6 + C_3H_8$

116. By Wurtz reaction, a mixture of methyliodide and ethyliodide gives
(a) Butane
(b) Ethane
(c) Propane
(d) A mixture of the above three

117. Product obtained by nitration of propane is
(a) Nitropropane
(b) Nitromethane
(c) Nitroethane
(d) All of these

118. Isomerism in saturated hydrocarbons is due to
(a) Change in the valence of carbon
(b) Change in the ratio of elements in compounds
(c) Formation of branches in the chain of C atoms
(d) Formation of double bond

119. Photochemical chlorination of alkane is initiated by a process of
(a) Pyrolysis
(b) Substitution
(c) Homolysis
(d) Peroxidation

120. Which of the following is not linked with methane
(a) Marsh gas
(b) Natural gas
(c) Producer gas
(d) Coal gas

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